

Total no. of Pages: 2

Roll no.

Degree: B-Tech.

Semester 3rd

END-SEMESTER EXAMINATION, June, 2026

Course Title: Probability and Stochastic Process
Course Code: ITECC06

Duration: 3.00 Hours
Max. Marks: 40

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																				
1.	Attempt any two parts of the following:																						
1a	Let X be a continuous random variable with the following distribution: $f(x) = \begin{cases} kx, & \text{if } 0 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$ (a) Evaluate k . (b) Find $P(1 \leq X \leq 3)$.	4	C01																				
1b	Let X and Y be independent random variables. Then show that (i) $E[XY] = E[X] E[Y]$, (ii) $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$.	4	C01																				
1c	Find the expectation (μ), variance (σ^2), and standard deviation (σ) of the following distribution <table border="1"><tr><td>x_i</td><td>2</td><td>3</td><td>11</td></tr><tr><td>$f(x_i)$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{2}$</td><td>$\frac{1}{8}$</td></tr></table>	x_i	2	3	11	$f(x_i)$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$	4	C01												
x_i	2	3	11																				
$f(x_i)$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$																				
2.	Attempt any two parts of the following:																						
2a	Let X and Y be random variables with joint distribution given below: <table border="1"><tr><td>$X \backslash Y$</td><td>-3</td><td>2</td><td>4</td><td>Sum</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.5</td></tr><tr><td>3</td><td>0.3</td><td>0.1</td><td>0.1</td><td>0.5</td></tr><tr><td>Sum</td><td>0.4</td><td>0.3</td><td>0.3</td><td></td></tr></table> Find: (a) the distributions of X and Y , (b) $\text{Cov}(X, Y)$	$X \backslash Y$	-3	2	4	Sum	1	0.1	0.2	0.2	0.5	3	0.3	0.1	0.1	0.5	Sum	0.4	0.3	0.3		4	C02
$X \backslash Y$	-3	2	4	Sum																			
1	0.1	0.2	0.2	0.5																			
3	0.3	0.1	0.1	0.5																			
Sum	0.4	0.3	0.3																				
2b	Team A has probability $2/3$ of winning whenever it plays. If A plays 4 games, find the probability that A wins (i) exactly 2 games, (ii) at least 1 game, (iii) more than half of the games.	4	C02																				
2c	Let X be a random variable with the Poisson distribution $p(k, \lambda)$. Then show that (i) $E(X) = \lambda$ and (ii) $\text{Var}(X) = \lambda$.	4	C02																				
3.	Attempt any two parts of the following:																						
3a	State and prove Chebyshev inequality	4	C03																				
3b	Let X be a random variable with the following distribution and let $Y = X^2$. Determine (i) the distribution (g) of Y (ii) the joint distribution (h) of X and Y .	4	C03																				

	<table><tr><td>x_i</td><td>-2</td><td>-1</td><td>1</td><td>2</td></tr><tr><td>$f(x_i)$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td></tr></table>	x_i	-2	-1	1	2	$f(x_i)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$		
x_i	-2	-1	1	2									
$f(x_i)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$									
3c.	The joint PDF of the random variables X and Y is given as $f_{XY}(x, y) = \frac{1}{4} e^{- x - y } \quad -\infty < x < \infty, -\infty < y < \infty$ Determine (i) Whether the random variables X and Y are statistically independent, (ii) The probability that $X \leq 1$ and $Y \leq 0$.	4	C03										
4.	Attempt any two parts of the following:												
4a	Define the following terms in detail: (i) Stationary random Process (ii) Ergodic process	4	C04										
4b	State and prove central limit theorem.	4	C04										
4c	If the random process is wide sense stationary than prove that $R_{XX}(-\tau) = R_{XX}(\tau)$	4	C04										
5.	Attempt any two parts of the following:												
5a	A man's smoking habits are as follows. If he smokes filter cigarettes one week, he switches to nonfilter cigarettes the next week with probability 0.2. On the other hand, if he smokes nonfilter cigarettes one week, there is a probability of 0.7 that he will smoke nonfilter cigarettes the next week as well. In the long run, how often does he smoke filter cigarettes?	4	C05										
5b	A gambler's luck follows a pattern. If he wins a game, the probability of winning the next game is 0.6. However, if he loses a game, the probability of losing the next game is 0.7. There is an even chance that the gambler wins the first game. (i) What is the probability that he wins the second game? (ii) What is the probability that he wins the third game? (iii) In the long run, how often will he win?	4	C05										
5c	For a Markov chain, the transition matrix is $P = \begin{pmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{pmatrix}$ with initial probability distribution $P^{(0)} = \left(\frac{1}{4}, \frac{3}{4}\right)$. Find (i) $P_{21}^{(2)}$, $P_{12}^{(2)}$ and $P^{(2)}$.	4	C05										